Impact Assessment Of Wind Farms On IEEE 14 Bus System For Small Signal Stability Analysis

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Abstract— **Stability is inherent characteristic of system, for efficient operation of system its ability to remain in normal state is desirable. In power systems usually the generation is done by thermal or hydro power plants, so as system dynamic and stability depends on these power plants stability analysis, where as in integration of renewable energy source incoming system dynamic capability with power systems to be analyzed and then allowed. To analyze the compatibility of SCIG and DFIG with the integration of power systems their modelling via PSAT to IEEE 14 bus test system is done in this research work. Time Domain analysis and eigen value analysis for stability study is done along with mathematical evaluation. Comparison of small signal stability between SCIG and DFIG is done.**

Keywords— Small Signal Stability; SCIG; DFIG; PSAT

I. INTRODUCTION

In this era renewable energy generation become prominent due to this wind energy installations were also increased [1]. Renewable energy is getting popularized with the introduction of distributed renewable energy generation and their integration with main grid through sub-grids or microgrids. Wind and Solar energy is mainly dependent on the atmospheric conditions [2], this is an issue to use this resources as base energy utility or say emergency power resources while can be used as peak load when base plants are working efficiently. Due to this reasons operations of wind power plant should be responsibly planned, so that incoming and outgoing machines in the system will not create disturbances.

Wind power farms produces active power where, as reactive power production should be done in reference to the power quality of the system to which this plants are providing power. Presently both type of wind power generation systems are utilized fixed speed and variable speed, hence to understand power quality issues are important to improvise the operations [4-6]. Integration of wind farms with large power pool will have inertia of machines which is to be managed, this leads to affect transient stability and small signal stability of system and wind farms. In this research work IEEE 14 bus system is modified with SCIG and DFIG based wind machines to investigate small signal stability. Modified system is utilized for investigation of different stability conditions by creating situations' and analyzing system behavior. All this simulation were done in PSAT 2.1.8 and simulation is analyzed in section IV and concluded in V.

II. SMALL SIGNAL STABILITY

Stability of power system is its ability to remain in synchronism after perturbation. In context to power systems classification of stability depends on its impacts to be studied, here small signal stability is analyzed which is rotor angle stability. Power system response in terms of state variables and S domain is analyzed here. Due to inability of machine to remain synchronism and maintain system dynamic, this leads to small signal stability.

In PSAT Eigen value and system dynamics correlations were analyzed to determine small signal stability. Where system state space and output equations are

$$X'(t) = f(x,u,t) \quad (1) \qquad\qquad y(t) = g(x,u,t) \quad (2)$$

Where, state variables: 'u 'input variables, 't' time and 'y 'output function.

As system is not linear to study the effect on system, linearization of equation (1) & (2) will lead to analyze small variations. Taylor's series is utilised to obtain polynomial equations in which higher order terms are omitted. Linear system is as:

$$X' = Ax + Bu \qquad (3)$$
$$y = Cx + Du$$

Here A, B, C and D are obtained from the Jacobean matrix which contains partial derivate of the functions in terms of 'f' and 'g' respectively to the input variable 'u' and the Where A , B , C and D are state matrix, input matrix, output matrix and coefficient matrix respectively [11].

$$A = \begin{bmatrix} \frac{df_1}{dx_1} & \cdots & \frac{df_1}{dx_n} \\ \cdots & \cdots & \cdots \\ \frac{df_n}{dx_1} & \cdots & \frac{df_n}{dx_n} \end{bmatrix} B = \begin{bmatrix} \frac{df_1}{du_1} & \cdots & \frac{df_n}{du_r} \\ \cdots & \cdots & \cdots \\ \frac{df_n}{du_1} & \cdots & \frac{df_n}{du_r} \end{bmatrix}$$

$$C = \begin{bmatrix} \frac{dg_1}{dx_1} & \cdots & \frac{dg_1}{dx_n} \\ \cdots & \cdots & \cdots \\ \frac{dg_m}{dx_1} & \cdots & \frac{dg_m}{dx_n} \end{bmatrix} D = \begin{bmatrix} \frac{dg_1}{du_1} & \cdots & \frac{dg_1}{du_r} \\ \cdots & \cdots & \cdots \\ \frac{dg_m}{du_1} & \cdots & \frac{dg_m}{du_r} \end{bmatrix}$$

$$\det(sI-A) = 0 \qquad (4)$$

A. SCIG Wind turbine modeling

In this research work PSAT tool box is provided inputs in terms of rotating reference frame. System is modeled in terms real (r) and imaginary (m) parameters, b_c is capacitor conductance; Equivalent circuit of SCIG is shown in figure 1.0. Where machine parameters are as:

$$v_r = V \sin(-\theta)$$
$$v_m = V \cos(\theta) \qquad (5)$$

And the absorbed power is given by

$$P = vrir + vmiqm$$
$$P = v_m i_r - v_r i_m + b_c (v_r^2 + v_m^2) \qquad (6)$$

Differential equations, of voltage for stator resistance r_s is as:

$$er' - vr = r_S ir - x\, i_m$$
$$e_m' - v_m = r_S i_m + x\, i_r \qquad (7)$$

The relation between currents, voltage and state variables is as follows:

$$er' = \Omega b(1-\omega m)em' - (e1r - (x0 - x')im)/T0'$$

$$em' = -\Omega_b (1-\omega_m) e_r' - (e_{1m} - (x_0 - x')i_r)/T_0' \qquad (8)$$

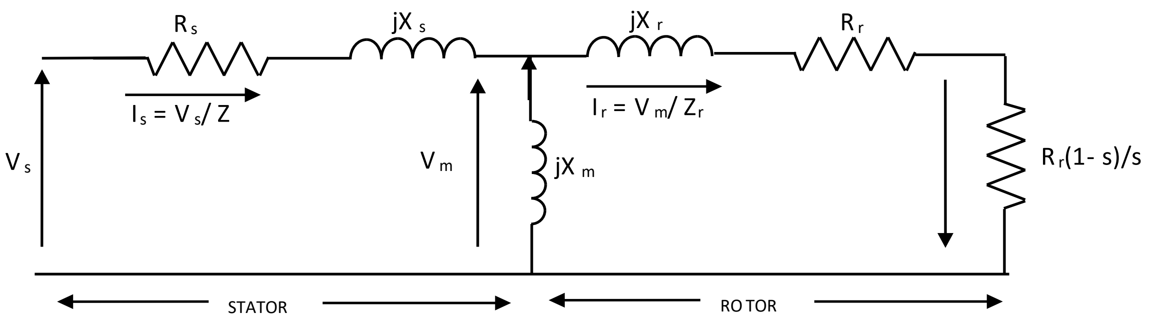


Figure 1.0: Equivalent Circuit Diagram of SCIG

Here x_0, x' and T_0 are calculated from the generator parameters.

$x_0 = x_s + x_m$
$x' = xS + xRxm/(x_R + x_m)$ (9)
$T_0 = (x_R + x_m)/\Omega_{brR}$

The differential equations in terms of mechanical parameters such as rotor inertia H_m, turbine inertia H_{wr} and shaft stiffness K_s is given by

$\omega wr = (T_{wr} - K_s\gamma)\ (/\ 2H_{wr})$

$\omega m = (Ks\gamma - Te)\ (/\ 2Hm)$ (10)

$\gamma = \Omega b(\omega wr - \omega m)$

Where Te (electrical torque) is given by

$T_e = {}_{er}'\ i_r + {}_{em}'\ i_m$ (11)

The mechanical torque is:

$Twr = Pw\ /\omega_{wr}$ (12)

Here Pw is the mechanical power extracted from wind. Wind and the rotor speeds are related as

$P_w = (\rho/2) c_p (\lambda)\ A_r v_w^3$ (13)

ρ= air density

Cp= Rotor power coefficient λ = Tip speed ratio A_r = Area of cross section of rotor v_w = Wind speed

B. DFIG Wind turbine modeling

Doubly fed induction generator (DFIG) is modeled via providing the mathematical modeling and parameters in PSAT. Equivalent circuit of DFIG is shown in Figure 2.0, these assumptions lead to

$$
\begin{aligned}
vds &= -rSids + ((xS + xm)iqs + xmiqr) \\
vqs &= -rSiqs - ((xS + xm)ids + xmidr) \qquad (14) \\
vdr &= -rRidr + (1-\omega m)((xR + xm)iqr + xmiqs) \\
vqr &= -rRiqr - (1-\omega m)((xR + xm)idr + xmids)
\end{aligned}
$$

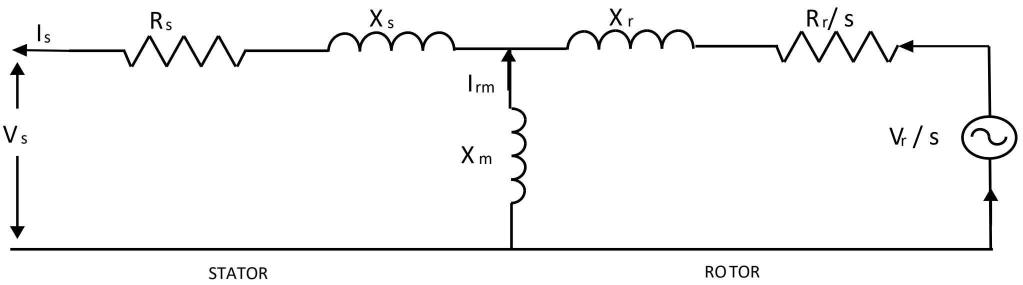


Figure 2.0 : Equivalent Circuit of DFIG

Stator voltage depends on both grid voltage magnitude and phase:

$$
\begin{aligned}
v_{ds} &= V \sin(-\theta) \\
v_{qs} &= V \cos(\theta) \qquad (15)
\end{aligned}
$$

The injected P & Q in to the grid depends on both stator current and grid side current of the converter. Equations are as follows:

$$
\begin{aligned}
P &= vdsids + vqsiqs + vdcidc + vqciqc \\
Q &= vqsids - vdsiqs + vqcidc - vdciqc \qquad (16)
\end{aligned}
$$

Grid side converter powers are:

$Pc = vdcidc + vqciqc$

$Qc = vqcidc - vdciqc$ (17)

Rotor side powers are

$Pr = vdridr + vqriqr$

$Q_r = v_{qr} i_{dr} - v_{dr} i_{qr}$ (18)

For loss less system final injected power is given as

P= $vdsids + vqsiqs + vdridr + vqriqr$ (19)

Q= $v_{qs} i_{ds} - v_{ds} i_{qs}$ (20)

In generator, motion equation single shaft model is used and it is assumed that converter controls can be able to filter shaft dynamics. For this reason, tower shadow effect is not considered. So

$\omega\square m = (T_m - T_e)/2H_m$ (21)

$T_e = \psi_{ds} i_{qs} - \psi_{qs} i_{ds}$

The relation between generator currents, stator flux is given as:

$\psi ds = -((xS + xm)ids + xmidr)$ $\psi_{qs} = -((x_S + {}_{xm})i_{qs} + {}_{xmiqr})$ (22)

Thus the electrical torque Te is given by

$T_e = x_m(i_{qr} i_{ds} - i_{dr} i_{qs})$ (23)

Converter currents differential equations are as follows:

$i_{dr} = K_V(V - V_{ref}) - V / x_m - i_{dr}$ (24)

Where $P_w^*(\omega_m)$is known as power – speed characteristic which approximately optimizes the wind energy capture and it is calculated based on current rotor speed value [12].

III. TEST SYSTEMS

In this IEEE 14 bus test system which is already existing is modified in bus number 3 SCIG and DFIG were connected to get desired analysis in this interconnected system. In this time domain simulation for fifty second were done with wind speed taken in PU for 15m/s as maximum wind speed, figure 3, shows IEEE 14 bus system.

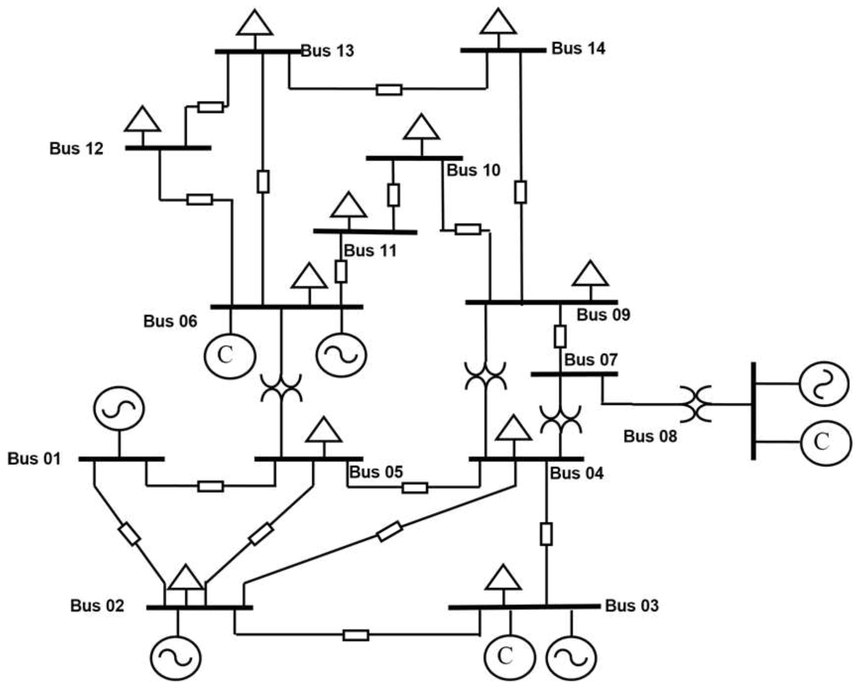
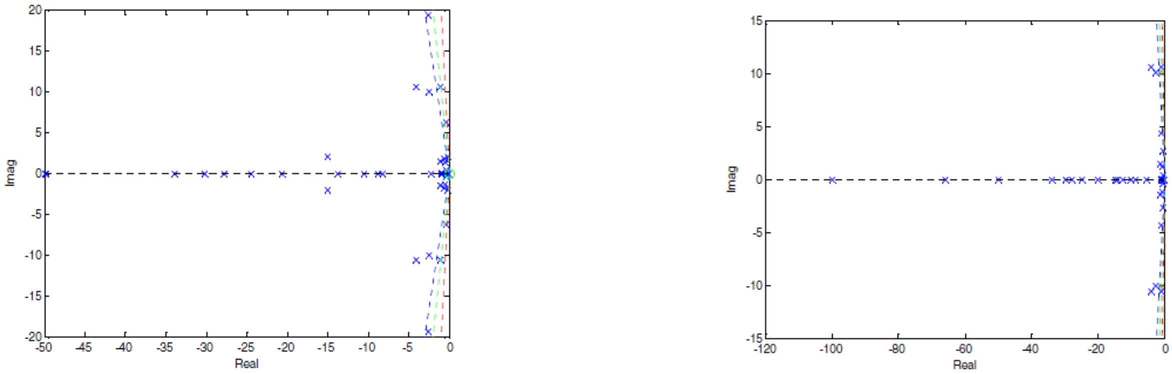


Figure 3.0 IEEE 14 Bus test system

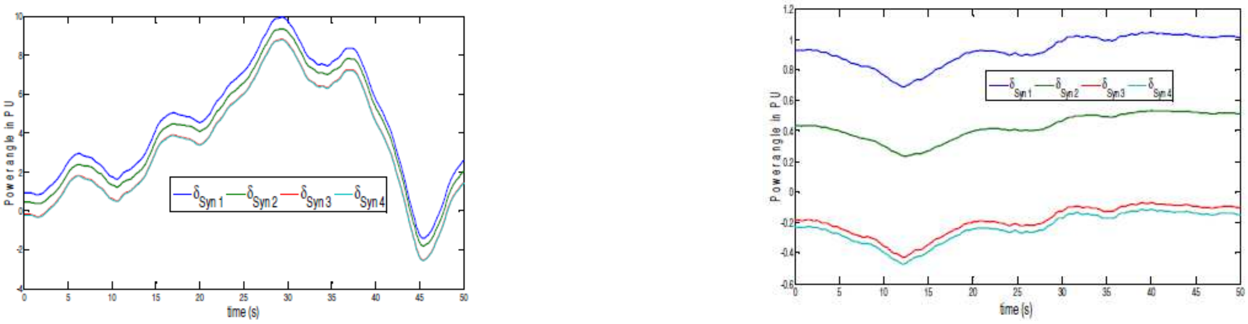
IV. SIMULATION RESULTS & DISCUSSIONS.

When integration of wind farms with power system occurs, due to inertia of wind turbine small signal stability problem were introduced into system, as this type of system are weather dependent and rotor angle parameters affects the systems dynamic. In this research work IEEE 14 bus system bus number 3 is associated with both DFIG and SCIG and response of both on the system were analyzed and compared. In SCIG as shown in figure 4 (a), all pole lies in left side of S- plane except one which is in origin where as in DFIG as shown in figure 4 (b) all poles are in left side of S-Plane thus the former one is marginally stable and later one is completely stable when small signal stability is considered.



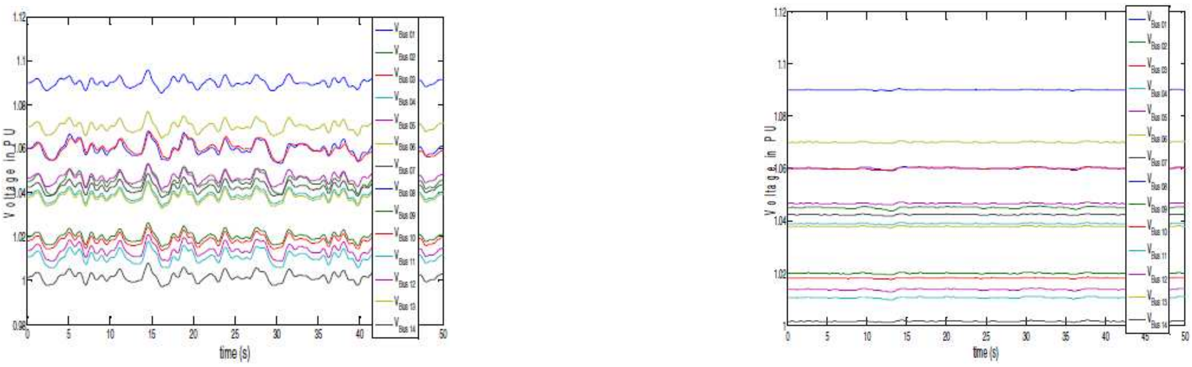
(a) (b)

Figure. 4 S Domain analysis of modified IEEE14 bus system with (a) SCIG (b) DFIG



(a) (b)

Figure. 5. Power angle deviations of modified IEEE14 bus system with (a) SCIG (b) DFIG



(a) (b)

Figure. 6. Voltage variations of modified IEEE14 bus system with (a) SCIG (b) DFIG

As shown in comparison of small signal stability by taking rotor angle stability and voltage stability for SCIG and DFIG in figure 5 and 6. SCIG having deviation in rotor angle and voltage stability when simulation were done whereas DFIG is having stable characteristic for the same system and response in IEEE 14 bus system. With this analysis small signal stability for SCIG and DFIG concludes that DFIG to be recommended for integration of wind farms into the power system as it have resilient characteristic.

V. CONCLUSION

In this research integration of wind farm with power system is analysied by modification of IEEE 14 bus system. As power system mostly have thermal or hydro power plants as the base load plant and system dynamics depends on this whereas integration of renewable energy system leads to many other issues to be resolved. To study the inertia of incoming machines of wind farms preliminary analysis were done in simulation, so as here small signal stability for SCIG and DFIG in 14 bus modified system. The result of this analysis after eigen value analysis along with stability test DFIG based wind generator is preferred over SCIG for integration of wind farms into grid. SCIG contain fixed shunt capacitor for the supply of reactive power, where as in DFIG grid side converter and rotor side converter feedback control the reactive power in the system. DFIG system is found more stable to be integrated into power systems due its resilient characteristics.

VI REFERENCES

[1] F. Grimaccia, M. Mussetta, R.E. Zich, "Neuro-fuzzy predictive model for PV energy production based on weather forecast," *2011 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE),* pp.24542457, 27-30 June 2011.

[2] M.Q. Duong, E. Ogliari, F. Grimaccia, S. Leva, M. Mussetta, "Hybrid model for hourly forecast of photovoltaic and wind power," 2013 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE), 7-10 July 2013.

[3] G. N. Sava, S. Costinas, N. Golovanov, S. Leva, and M. Q. Duong, "Comparison of active crowbar protection schemes for DFIGs wind turbines," in 2014 IEEE 16th International Conference on Harmonics and Quality of Power, pp. 448-552, 2014.

[4] M. Q. Duong, K. H. Le, F. Grimaccia, S. Leva, M. Mussetta, and R. Zich, "Comparison of power quality in different grid-integrated wind turbines," in 2014 IEEE 16th International Conference on Harmonics and Quality of Power, pp. 669-673, 2014.

[5] M. Q. Duong, F. Grimaccia, S. Leva, M. Mussetta, E. Ogliari, "Pitch angle control using hybrid controller for all operating regions of SCIG wind turbine system," Renewable Energy, Elsevier, Vol. 70, pp. 197203, 2014.

[6] M. Q. Duong, F. Grimaccia, S. Leva, M. Mussetta, and R. Zich, "Improving LVRT characteristics in variable-speed wind power generation by means of fuzzy logics," in 2014 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE), pp. 332-337, July 2014.

[7] H. Schwarz, K. Pfeiffer, and L. Roskoden, "Integration of renewable eneries to the east German grid. Actual problems and possibe solutions," in Proc. Of the 6th WSEAS International Conference on Power Systems, pp. 7-13, 2006.

[8] A. G. Pillai, P. Thomas, K. Sreerenjini, S. Baby, T. Joseph, and S. Srecdharan, "Transient stability analysis of wind integrated power systems with storage using central area controller," in 2013 Annual International Conference on Emerging Research Areas and 2013 International Conference on Microelectronics, Communications and Renewable Energy, pp. 1-5, 2013.

[9] Yanhua Liu and Yongning Chi and Weisheng Wang and Huizhu Dai, "Impact of large scale wind farm integration on power system transient stability," in 2011 4th International Conference on Electric Utility Deregulation and Restructuring and Power Technolgies, pp. 1301-1305, 2011.

[10] Y. Chompoobutrgool, and Wei Li and L. Vanfretti, "Development and implementation of a Nordic grid model for Power System small-signal and transient stability studies in a free and open source software," in 2012 IEEE Power and Energy Society General Meeting , pp. 1-8, 2012.

[11] Y.U. Lopez, J.A.D. Navarro, "Small Signal Stability analysis of wind turbines with squirrel cage induction generators," IEEE International Conference on Transmission and Distribution Conference and Exposition pp.1-10, 2008.

[12] F. Milano, "Power System Analysis Toolbox," Version 1.3.4, Software and Documentation, July 14, 2005.

[13] Xie D., Lu Y., Sun J., ET AL.: 'Small signal stability analysis for different types of PMSGs connected to the grid', Renew. Energy, 2017, 106, pp. 149–164

[14] Ibrahim R.A., Hamad M.S., Dessouky Y.G., ET AL.: 'A review on recent low voltage ride-through solutions for PMSG wind turbine'.IEEE 2012 Int. Symp. Power Electronics, Electrical Drives, Automation and Motion (SPEEDAM), Sorrento, Italy, 2012

[15] Chiniforoosh S., Jatskevich J., Yazdani A., ET AL.: 'Definitions and applications of dynamic average models for analysis of power systems', IEEE Trans. Power Deliv., 2010, 25, (4), pp. 2655–2669

[16] Sun J.: 'Unified averaged switch models for stability analysis of large distributed power systems'. IEEE Fifteenth Annual Applied Power Electronics Conf. Exposition (APEC 2000), New Orleans, LA, USA, 2000, vol. 1

[17] Gagnon R., Sybille G., Bernard S., ET AL.: 'Modeling and real-time simulation of a doubly-fed induction generator driven by a wind turbine'. Int. Conf. Power Systems Transients, Montréal, Canada, 2005

[18] Dommel H.W.: 'EMTP theory book' (Microtran Power System Analysis Corporation, Vancouver, British Columbia, 1996)

[19] F. Mei and B. C. Pal, "Modal Analysis of Grid Connected Doubly Fed Induction Generator," *IEEE Transactions on Energy Conversion*. Vol. 22, No. 3, 2007, pp. 728-736. doi:10.1109/TEC.2006.881080

[20] F. Mei and B. C. Pal, "Modeling and Small Signal Analysis of a Grid Connected Doubly Fed Induction Generator," *IEEE Power Engineering Society General Meeting*, San Francisco, 12-16 June 2005,pp. 2101-2108.

[21] F. Wu, X. Zhang, K. Godfrey and P. Ju, "Modeling and Control of Wind Turbine with Doubly Fed Induction Generator" *2006 IEEE PES Power System Conference and Exposition*, 29 October-1 November 2006, pp. 1404-1409. doi:10.1109/PSCE.2006.296507

[22] P. Kundur, "Power System Stability and Control" McGraw-Hill, Inc., New York, 19

[23] Anurag S D Rai, Reeta Pawar, Durga Sharma, Shaurabh Sen, Sanjeev Kumar Gupta; Algorithms For Synchrophasor Enabled Digital Relay In Differential Protection Scheme, Proceedings Of International Conference On Recent Advancement On Computer And Communication, Lecture Notes In Networks And Systems 34, https://doi.org/10.1007/978-981-10-8198-9_4 , Book Id: 448040_1_En, Book ISBN: 978-981-10-8197-22018, Springer Nature Singapore Pte Ltd.

[24] Anurag S D Rai, Reeta Pawar, Anil Kurchania, C S Rajeshwari; Cyber Physical Systems And Resilience In Micro Grid, Anusandhan, Rabindranath Tagore University, Volume:7, Issue:14, Month: Sep, 2018. http://aujournals.ipublisher.in/I/a/57822

[25] Reeta Pawar, Anurag S D Rai, Anil Kurchania, Alpana Pandey; A Review On Different Types Of Wind Generation, Anusandhan, Rabindranath Tagore University, Volume:7, Issue:14, Month: Sep, 2018. http://aujournals.ipublisher.in/I/a/57827